Task	Status	Action
Phase 1: LangChain Fundamentals		
LangChain Basics & Setup		
Understanding LangChain Components (Tools, Agents, Chains)		Explore more (use different tools)(how agents work)
Hands-on: Running Simple LangChain Chains		
Integrating LangChain with LLMs (OpenAI, Ollama)		
LangChain Output Parsers & Memory		
Hands-on: LangChain with Prompt Templates		
Debugging and Logging in LangChain		
Phase 2: RAG (Retrieval-Augmented Generation)		
Understanding Embeddings & Vector Databases		
Deep Dive into Vector Search		
Hands-on: Creating First Embeddings		
Implement FAISS/ChromaDB Storage		1
Testing Retrieval of Stored Embeddings		4
Understanding Document Parsing		2
Text Extraction from PDFs/Word		
Implementing Chunking & Summarization		
Preprocessing Strategies for Better Retrieval		
Hands-on: Implementing Text Chunking		3
Enhancing Retrieval Strategies in RAG		
Hybrid Search: Combining Keyword & Vector Search		
Implementing Ranking & Reranking with BM25		
Testing Retrieval Results		
Hands-on: Improving Query Responses		
Phase 3: Building & Testing the RAG Pipeline		
Building a Basic RAG Pipeline		
Hands-on: Querying the RAG Model		
Chatbot Integration with RAG		
Hands-on: Testing Chatbot Conversations		

Phase 4: Real-Time Chatbot Responses with Sockets		
Understanding Streaming in Chatbots		
Intro to WebSockets & Socket.IO		
Implementing Streaming Responses in Chatbot		
Hands-on: Chatbot with Real-Time Output		
Phase 5: Speech Processing (STT + TTS)		
Speech Processing: Overview of STT & TTS		
Using OpenAI Whisper for STT		
Understanding TTS Models		
Hands-on: Implementing Basic TTS		
Phase 6: Voice-Enabled RAG Chatbot Integration		
Implementing STT for Voice Queries		
Exploring Speaker Diarization		
Integrating STT with Chatbot		
Hands-on: Running Chatbot with Voice Queries		
Phase 7: Final Optimization, Deployment & Demo		
Optimizing Chatbot Responses		
Understanding Security & Privacy in AI		
Mitigating Biases in AI Models		
Preparing Chatbot for Deployment		
API Security & Rate Limiting		
Deploying with Docker & Kubernetes		
Presenting Final Project Demo		